1 SHEAR

1.1 CONCRETE CONTRIBUTION V_c

Primary (ACI 318 simplified):

$$V_c = 2\lambda\sqrt{f'_c}\, b_w\, d \tag{1}$$

λ lightweight concrete modification factor, 1.0 normal weight
f'_c specified concrete compressive strength [psi]
b_w web width [in]
d effective depth to tension steel centroid [in]

LIMITS

Stirrups required	when $V_u > 0.5\,\phi V_c$

1.2 STEEL CONTRIBUTION V_s

$$V_s = \frac{A_v\, f_{yt}\, d}{s} \tag{2}$$

A_v area of shear reinforcement within spacing s [in^2]
f_{yt} yield strength of transverse reinforcement [psi]
s stirrup spacing along member axis [in]

LIMITS

Upper cap on V_s	$V_s \le 8\sqrt{f'_c}\, b_w d$
Min stirrup area	$A_{v,\min} = \max\left(0.75\sqrt{f'_c}\,\dfrac{b_w s}{f_{yt}},\ \dfrac{50\, b_w s}{f_{yt}}\right)$
Max spacing, $V_s \le 4\sqrt{f'_c}\, b_w d$	$s_{\max} = \min(d/2,\ 24\ \text{in})$
Max spacing, $V_s > 4\sqrt{f'_c}\, b_w d$	$s_{\max} = \min(d/4,\ 12\ \text{in})$

1.3 COMBINED

$$\phi V_n = \phi(V_c + V_s) \ge V_u, \qquad \phi = 0.75 \tag{3}$$

2 MOMENT

2.1 STRESS BLOCK DEPTH a

$$a = \frac{f_y A_s}{0.85 f'_c b} \tag{4}$$

f_y yield strength of longitudinal tension reinforcement [psi]
A_s area of longitudinal tension reinforcement [in^2]
f'_c specified concrete compressive strength [psi]
b width of compression face [in]

2.2 NOMINAL MOMENT CAPACITY

$$\phi M_n = \phi f_y A_s \left(d - \frac{a}{2}\right), \qquad \phi = 0.9 \tag{5}$$

2.3 REQUIRED STEEL AREA $A_{s,\text{req}}$

Combining a and $\phi M_n \ge M_u$ and solving the quadratic in a:

$$A_{s,\text{req}} = \frac{0.85 f'_c b}{f_y}\left[d - \sqrt{d^2 - \frac{2M_u}{0.85\,\phi\, f'_c\, b}}\right] \tag{6}$$

M_u factored moment demand [lb-in]

3 DEFLECTION

3.1 CRACKING MOMENT M_{cr}

$$M_{cr} = \frac{f_r\, I_g}{y_t}, \qquad f_r = 7.5\lambda\sqrt{f'_c} \tag{7}$$

f_r modulus of rupture [psi]
I_g gross (uncracked) moment of inertia [in^4]
y_t distance from centroid to extreme tension fiber [in]

3.2 CRACKED TRANSFORMED SECTION I_{cr}

Modular ratio and concrete modulus:

$$n = \frac{E_s}{E_c}, \qquad E_c = 57{,}000\sqrt{f'_c}\ \text{[psi]} \tag{8}$$

Neutral axis depth kd from first moments about the NA (tension concrete ignored):

$$\frac{b(kd)^2}{2} = nA_s(d - kd) \tag{9}$$

Cracked transformed moment of inertia:

$$I_{cr} = \frac{b(kd)^3}{3} + nA_s(d - kd)^2 \tag{10}$$

n modular ratio, steel to concrete
E_s steel modulus of elasticity, typically 29,000 ksi
E_c concrete modulus of elasticity [psi]
kd depth from extreme compression fiber to neutral axis [in]

3.3 EFFECTIVE MOMENT OF INERTIA I_e (ACI 318-19 §24.2.3.5)

Section is uncracked while $M_a \le \frac{2}{3}M_{cr}$:

$$I_e = I_g \tag{11}$$

Once $M_a > \frac{2}{3}M_{cr}$, use the Bischoff form:

$$I_e = \frac{I_{cr}}{1 - \left(\frac{2}{3}\frac{M_{cr}}{M_a}\right)^2\left(1 - \frac{I_{cr}}{I_g}\right)} \tag{12}$$

M_a maximum service moment at the stage deflection is computed [lb-in]

LIMITS

Upper cap	$I_e \le I_g$
Asymptote	$I_e \to I_{cr}$ as $M_a \to \infty$

3.4 PROCEDURE FOR I_e

1. Compute $I_g = bh^3/12$ and $y_t = h/2$.
2. Compute $f_r = 7.5\lambda\sqrt{f'_c}$ and $M_{cr} = f_r I_g / y_t$.
3. If $M_a \le \frac{2}{3}M_{cr}$, set $I_e = I_g$ and stop.
4. Compute $E_c = 57{,}000\sqrt{f'_c}$ and $n = E_s/E_c$.
5. Solve the cracked-section quadratic for kd, then $I_{cr} = b(kd)^3/3 + nA_s(d - kd)^2$.
6. Apply Bischoff to get I_e, then cap at I_g.
7. $EI_{\text{eff}} = E_c I_e$ for the deflection calculation.